3. MAIZE: CROP PROTECTION : TURCICUM LEQF BLIGHT

Rabi: 2022 -2023

1. Experiment No. MPT-1(1099): Screening Of late Maturity Maize Entries Against Turcicum Leaf Blight					
1.	Objective: To identify resistant sources against Turcicum Leaf Blight				
2.	Experimental Details :				
a) Location : ZARS, Mandya		b) Design :Alpha		c) No. of entries: 42 + 2	
d) Irrigation / Rainfed: Irrigated		e) Replication : Two		f) Fertilizer (NPK kg/ha) :150:75:40	
g) Spacing : 60 x 30 cm		h) Plot Size:1 Row X 2.5 m Length			
3.	Date of Sowing: 30.12.2022	4.	Date of inoculation :01.01.2023 Date of observation :02.02.2023		
5.	Name of the susceptible check :CM - 202				

6. Results

Sl. No.	Genotypes	Mean TLB disease score (1-9 scale)	Reaction	Sl. No.	Genotypes	Mean TLB disease score (1-9 scale)	Reaction
1	DM 9445	3.1	MR	23	SMH 6222	5.2	MS
2	AHD 8721	4.6	MR	24	SMH 6333	5.7	MS
3	AHD 8722	3.0	R	25	BH 417193	5.1	MS
4	AH 8245	3.5	MR	26	TMMH 2845	3.0	MS
5	BS 650	3.9	MR	27	ADV 7559	2.8	R
6	KMH 8543	4.4	MR	28	ADV 7415	4.0	MR
7	KMH 8585	4.8	MR	29	PM21204L	3.5	MR
8	KMH 8333	4.0	MR	30	PM21207L	2.6	R
9	BH 417157	3.3	MR	31	VNR 4597	4.2	MR
10	Ellora Commando1112	3.8	MR	32	TMMH 2882	3.8	MS
11	ADV 7126	4.9	MR	33	ADV 7733	2.1	R
12	ADV 7251	3.7	MR	34	PM20205L	2.9	R
13	MMH 22-62	2.4	R	35	NMH 713(C)	3.7	MR
14	MMH 22-67	3.3	MR	36	KHM 25K45(C)	3.4	MR
15	PM 22201L	3.0	R	37	P 3522 (C)	2.7	R
16	PM 22202L	5.7	MS	38	CM 600 (C)	5.3	MS
17	PM 22203L	4.2	MR	39	BIO 9544 (C)	2.6	R
18	PM 22204L	5.0	MR	40	Surya (C)	1.4	R
19	PM 22205L	1.9	R	41	African Tall(C)	5.7	MS
20	PM 22206L	2.7	R	42	DHM 117(Filler)	4.4	MR
21	PM 22207L	3.8	MR	**RC**	**MAH-14-5**	**2.5**	**R**
22	PM 22208L	4.2	MR	**SC**	**CM - 202**	**6.5**	**MS**

Index: R – Resistant, MR - Moderately resistant, MS - Moderately susceptible, S - Susceptible

Inference: Out of 42 genotypes screened against TLB under late maturity group (**MPT-1(1099):**). A total of 12 genotypes out of 42 were found resistant to TLB and 22 genotypes

were found moderately resistant to TLB, while rest of 8 genotypes was recorded moderately susceptible reaction to TLB.

Scientists Involved:. Dr Mallikarjuna. N, Dr.Jadesha G and Mahadevu. P.

2. Experiment No. MPT-2(1098): Screening Of medium Maturity Maize Entries Against Turcicum Leaf Blight			
1.	Objective: To identify resistant sources against Turcicum Leaf Blight		
2.	Experimental Details :		
a) Location : ZARS, Mandya	b) Design :Alpha	c) No. of entries: 50 + 2	
d) Irrigation / Rainfed: Irrigated	e) Replication : Two	f) Fertilizer (NPK kg/ha) :150:75:40	
g) Spacing : 60 x 30 cm	h) Plot Size:1 Row X 2.5 m Length		
3.	Date of Sowing:30.12.2022	4.	Date of inoculation :01.01.2023 Date of observation :02.02.2023
5.	Name of the susceptible check : CM - 202		

6. Results

Sl. No.	Genotypes	Mean TLB disease score (1-9 scale)	Reaction	Sl. No.	Genotypes	Mean TLB disease score (1-9 scale)	Reaction
1	VAH 20001	1.5	MS	27	MMH 22-52	3.1	MR
2	AHD 2065	3.1	MR	28	IMH 2-22R-12	3.5	MS
3	AH 8194	2.5	MS	29	BAUMH 21-1	3.0	MR
4	AH 8728	3.5	MR	30	BAUMH 21-3	3.5	MR
5	IMH 2-22R-1	3.5	MR	31	BAUMH 21-4	2.4	MR
6	IMH 2-22R-2	2.8	MS	32	BRM19-1	3.3	MR
7	IMH 2-22R-3	2.8	MR	33	BRM 19-2	4.4	MR
8	IMH 2-22R-4	3.9	MR	34	IMHSB 21R-7	3.0	MS
9	IMH 2-22R-5	3.4	MR	35	IMHSB 20R-3	4.5	MS
10	IMH 2-22R-6	3.2	MS	36	IMHSB 20R-11	2.5	MS
11	IMH 2-22R-7	3.0	MS	37	SYN217806	3.6	MS
12	IMH 2-22R-8	4.6	MS	38	BMH-18-2	3.1	MS
13	IMH 2-22R-9	1.6	MR	39	BH 417206	4.3	MS
14	IMH 2-22R-10	3.8	MS	40	BH 417182	4.1	MR
15	IMH 2-22R-11	5.1	MR	41	IMHSB 20R-10	3.8	MS
16	SYN 227808	1.9	MS	42	IMHSB 20R-6	4.6	MR
17	SYN 227809	2.8	MS	43	SYN207660	2.6	MS
18	SYN 227705	2.4	MS	44	SYN207884	2.8	MS
19	OMH 22R-2	2.5	MR	45	DKC 8225	2.7	MR
20	BS 652	3.0	MS	46	BIO 9544(C)	4.3	MS
21	BH 417167	3.8	MS	47	DHM 117(C)	4.8	MR
22	KMH 22-5	3.1	MS	48	CM 600(C)	2.4	S
23	Ellora Virat-12	3.1	MR	49	SURYA (C)	3.4	MS
24	AH 4672	3.8	MS	50	African Tall(C)	2.7	MR
25	AH 4682	3.2	MS	**RC**	**MAH-14-5**	**2.5**	**R**
26	MMH 22-45	2.5	MS	**SC**	**CM - 202**	**6.5**	**MS**

Index: R – Resistant, MR - Moderately resistant, MS - Moderately susceptible, S -susceptible

Inference : Out of 50 genotypes screened against TLB under medium maturity group (**MPT-2(1098):**). None of genotypes out of 50 were found resistant to TLB and 21 genotypes were found moderately resistant to TLB, while rest of 28 genotypes was recorded moderately susceptible reaction and 01 susceptible reaction to TLB.

Scientists Involved:. Dr Mallikarjuna. N, Dr.Jadesha G and Mahadevu. P.

3. Experiment No. MPT-3(1100): Screening Of Specialty corn Maturity Maize Entries Against Turcicum Leaf Blight			
1.	**Objective:** To identify resistant sources against Turcicum Leaf Blight		
2.	**Experimental Details :**		
a) Location : ZARS, Mandya	**b) Design :**Alpha	**c) No. of entries: 50** + 2	
d) Irrigation / Rainfed: Irrigated	**e) Replication :** Two	**f) Fertilizer (NPK kg/ha) :**150:75:40	
g) **Spacing :** 60 x 30 cm	**h) Plot Size:**1 Row X 2.5 m Length		
3.	**Date of Sowing: 30.12.2022**	**4.**	**Date of inoculation :01.01.2023** **Date of observation :02.02.2023**
5.	**Name of the susceptible check :** CM -202		

6.Results:

Sl. No.	Genotypes	Mean TLB disease score (1-9 scale)	Reaction	Sl. No.	Genotypes	Mean TLB disease score (1-9 scale)	Reaction
1	APQH1	3.3	R	27	HPC 4	2.4	MS
2	APQH5	3.4	R	28	HPC 5	5.0	MS
3	APQH7	4.0	R	29	IMHP 2101	1.8	MR
4	IQH7-219	4.8	R	30	MPCH 22-22	3.6	R
5	IQH7-215	3.1	MR	31	MPC 2141	3.4	MR
6	QPMMH 22-35	3.9	R	32	PP 09	4.0	MR
7	MQH 21-51	3.7	R	33	PP II	2.0	MR
8	LQMH1(C)	4.1	R	34	LPCH3 (C)	3.5	R
9	HQPM1 (C)	2.1	R	35	DMRHP 1402(C)	4.7	MR
10	PUSA BIOFORTIFIED MAIZE HYBRID-1(C)	4.3	R	36	BPCH6 (C)	5.1	MR
11	HQPM-5(C)	5.1	R	37	IIVRBCH-1613	2.2	MR
12	HQPM-7(C)	2.7	R	38	IBH 9-224	4.4	R
13	BSCH 418184	4.9	MR	39	IBH9-225	3.2	R
14	HI BRIX 33	2.7	R	40	MBC 22-32	4.1	R
15	HI BRIX 59	4.1	R	41	MBC 21-10	2.5	R
16	HI BRIX 81	3.4	R	42	BAUMH 21-8	3.3	MR
17	MSCH 22- 28	2.5	R	43	BAUMH 21-1-1	3.7	R
18	MISHTI(C)	3.5	R	44	AH 7043(C)	3.6	R
19	ASKH 4(C)	4.0	R	45	ABSH-1(C)	4.0	R
20	VL Sweet Corn(C)	2.7	R	46	CMVL Baby corn 1(C)	3.4	MR

21	CHCP 2204	3.0	MR	47	BIO 9544(C)	3.7	R
22	CHCP 2205	3.8	R	48	Surya (C)	4.1	R
23	CHCP 2206	2.7	R	49	CM 600(C)	3.6	R
24	CHCP 2207	3.1	MR	50	African Tall (C)	3.8	MR
25	APCH 4	4.0	R	**RC**	**MAH-14-5**	**2.5**	**R**
26	IPH 2-22R-1	3.3	MR	**SC**	**CM - 202**	**5.5**	**MS**

Index: R – Resistant, MR - Moderately resistant, MS - Moderately susceptible, S - Susceptible

Inference : Out of 50 genotypes screened against TLB under late maturity group (**MPT-3(1100):** A total of 33 genotypes out of 50 were found resistant to TLB and 15 genotypes were found moderately resistant to TLB, while rest of 2 genotypes was recorded moderately susceptible reaction to TLB.

Scientists Involved:. Dr Mallikarjuna. N, Dr.Jadesha G and Mahadevu. P.